Name:

Collaborators:

Instructions: Worksheets are graded mostly on completion and partially on correctness. Please write complete solutions showing explanations and key steps to the following problems, unless it says otherwise.

Find Eigensolutions with Real Repeated Eigenvalues

1. The Independent Solution

Consider the homogeneous linear system of 1st-order ODEs: $\begin{aligned} x' &= y \\ y' &= -2\ x - 2\sqrt{2}\ y \end{aligned}$.

This system has real repeated eigenvalues with geometric multiplicity less than the algebraic multiplicity.

a. Show that the eigenvalue is $\lambda = -\sqrt{2}$ with algebraic multiplicity of 2.

b. Show that the corresponding eigenvector of λ is $\vec{v} = \begin{bmatrix} 1 \\ -\sqrt{2} \end{bmatrix}$ with geometric multiplicity of 1.

c. Based solely on the eigenvalue, what does this tell about the behavior of solutions around the equilibrium?

d. Let $\vec{w}_1 = \begin{bmatrix} x \\ y \end{bmatrix}$. The independent solution is

$$\vec{w}_1 = \vec{v}e^{(\lambda\ t)}.$$

where λ is the eigenvalue with its corresponding eigenvector $\vec{v}$.

i. Explain why specific solutions can only be found when the initial conditions lie on the line $y = -\sqrt{2}\ x$.

ii. Explain why $\vec{w}_1$ is not the general solution.

2. The General Solution with Real Repeated Eigenvalues

Consider the same linear system from Problem (1) in matrix-vector form: $\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -2 & -2\sqrt{2} \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$. The eigenvalue is $\lambda = -\sqrt{2}$ with algebraic multiplicity of 2 and the corresponding eigenvector is $\vec{v} = \begin{bmatrix} 1 \\ -\sqrt{2} \end{bmatrix}$ with geometric multiplicity of 1. This creates one linearly independent solution of the form

$$\vec{w}_1 = \vec{v}e^{(\lambda\ t)}.$$

In order to find the general solution, we need to find another linearly independent solution of the form

$$\vec{w}_2 = (t\ \vec{v} + \vec{u})\, e^{(\lambda\ t)}$$

where $\vec{u}$ is the *generalized eigenvector*.

a. Given the eigenvector $\vec{v} = \begin{bmatrix} 1 \\ -\sqrt{2} \end{bmatrix}$, we need to find the generalized eigenvector $\vec{u}$. The following tasks shows you how to determine $\vec{u}$ directly from the coefficient matrix.

Task	Show work here
Set $\left(\begin{bmatrix} 0 & 1 \\ -2 & -2\sqrt{2} \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \right) \vec{u} = \vec{v}$.	
Write the equation into an augmented matrix: $\begin{array}{l} \text{row}_1 \rightarrow \\ \text{row}_2 \rightarrow \end{array} \begin{bmatrix} -\lambda & 1 & 1 \\ -2 & -2\sqrt{2} - \lambda & -\sqrt{2} \end{bmatrix}$	
Row reduce the augmented matrix into the form $\begin{array}{l} \text{row}_1 \rightarrow \\ \text{row}_2 \rightarrow \end{array} \begin{bmatrix} a & b & c \\ 0 & 0 & 0 \end{bmatrix}$ where a, b, and c are values left over.	
The values in row_1 correspond to the coefficients of the line $ax + by = c$. Write this line in parametric vector form with s as the independent variable.	
To determine the vector $\vec{u}$, set $s = 0$.	

Now, you have two linearly independent solutions $\vec{w}_1$ and $\vec{w}_2$ which forms a fundamental set of solutions.

b. Use the superposition principle to write the general solution.

c. Determine the specific solution using the initial condition $(1, 0)$.